Q3 (a).

Given $a, b \in \mathbb{Z}$ and a ~~non-zero~~ positive modulus $n \in \mathbb{Z}$, $n > 0$.

we say "a is congruent to b modulo n", written as $a \equiv b \pmod{n}$.

if and only if n divides $a-b$,

ie. $n \mid a-b$.

and if not we can write.

$$a \not\equiv b \pmod{n}.$$

When a, b are divided by n leaving smallest ~~posi~~ non-negative remainders say

$$a = q_1 n + r_1$$
$$b = q_2 n + r_2$$

for some $q_1, q_2 \in \mathbb{Z}$, and remainders $r_1, r_2 \in \mathbb{Z}$. satisfying

$0 \le r_1, r_2 < n$

then

$$a \equiv b \pmod{n} \iff r_1 = r_2.$$

using proof by contradiction.

(b) Let $d = \gcd(a, n)$. Assume that $d \nmid b$, and there is an integer solution x to

$$ax \equiv b \pmod{n}.$$

$\iff n \mid ax - b$, by def of $\equiv$

$\iff \exists q \in \mathbb{Z} \quad ax - b = qn.$
, by def. of divisibility

$\iff \exists q \in \mathbb{Z} \quad b = ax - qn$

$\implies d \mid b$, since $d \mid a$ & $d \mid n$ and so d divides any linear combination of a, n.

This contradicts our assumption that $d \nmid b$, so therefore there are

$\boxed{\text{no solutions to } ax \equiv b \pmod{n}}$

(c) We need to find r, where $0 \le r < 17$ and

$$2025^{2026} \equiv r \pmod{17}.$$

$$2025^{2026} \equiv \underline{\quad 2 \quad}^{2026} \pmod{17}$$

, since $2025 \equiv 2 \pmod{17}$.

so $\forall e$ $2025^{e} \equiv 2^{e} \pmod{17}$.

17 is prime and Fermat's Little Theorem.

$$2^{16} \equiv 1 \pmod{17}$$

$$\begin{array}{r} 119 \\ 17\overline{)2025} \\ 17 \\ \hline 32 \\ 17 \\ \hline 155 \\ 153 \\ \hline 2 \end{array}$$

$2026 = 126 \cdot 16 + 10$

$$\begin{array}{r} 126 \\ 16\overline{)2026} \\ 16 \\ \hline 42 \\ 32 \\ \hline 106 \\ 96 \end{array}$$

So

$$2^{2026} = 2^{16 \cdot 126 + 10}$$

$$= (2^{16})^{126}\, 2^{10}, \text{ rules of exponents}$$

$$2025^{2026} \equiv 1^{126} \cdot 2^{10} \pmod{17}$$

$$\equiv 2^{10} \pmod{17}.$$

Evaluate this using repeated squaring. $(10 = 8+2 = 2^3 + 2^1)$

$$2^4 = 16, \quad 2^8 \equiv 16^2 \equiv (-1)^2 \pmod{17}$$

$$\equiv 1$$

So $2^{10} \equiv 2^8 \cdot 2^2 \pmod{17}$

$$\equiv 1 \cdot 4 \pmod{17}$$

$$\equiv 4 \pmod{17}.$$

So in summary

$$2025^{2026} \equiv 4 \pmod{17}$$

So the required remainder is 4.

Q5. Let $(G, *)$ satisfy.

$\forall g_1, g_2 \in G \quad g_1 * g_2 \in G$.

$*$ must be associative on G

ie. $\forall g_1, g_2, g_3 \in G$

$$(g_1 * g_2) * g_3$$
$$= g_1 * (g_2 * g_3)$$

$(G, *)$ must contain an identity element, ie.

$\exists e \in G \ \forall g \in G \quad g * e = e * g = g$

$(G, *)$ must contain inverses for all its elements, ie.

$\forall g \in G \ \exists g^{-1} \in G \quad g * g^{-1} = g^{-1} * g = e$.

So in $(\mathbb{R}, +)$ addition is associative for instance

$$-10 + (16 - 17) = -10 + (-1) = -11$$

$(-10+16)-17 = 6-17 = -11$

The identity for $(\mathbb{R},+)$ is 0.

as $6+0=6=0+6$ for example.

Given a $x \in \mathbb{R}$ its inverse, using additive notation, $-x$ is a real number also.

e.g. $17+(-17)=0$.

(b) $(\mathbb{Q},\times)$ does not form a group since $0 \in \mathbb{Q}$ does not have a multiplicative inverse as $\forall q \in \mathbb{Q}$ $0 \times q = 0 \neq 1$, the identity of $(\mathbb{Q},\times)$.

Exclude zero to form the set

$\mathbb{Q}^{\times} = \mathbb{Q} \setminus \{0\}$, which now form a group under $\times$, as the inverses

condition and others are now satisfied.

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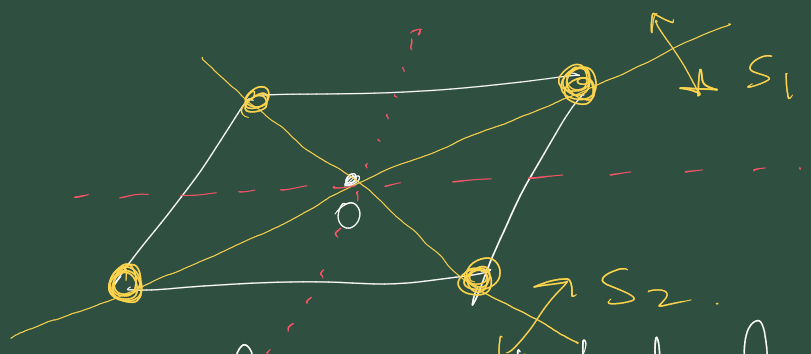
(c). Consider the rhombus R.



This has a total of four symmetries.

- The identity e.
- The rotation r, by π radians about origin.
- Two reflections s_1, s_2 in the diagonal lines joining opposite vertices.

$$\mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

~~Let~~

Let $G_R = \{e, r, s_1, s_2\}$

this forms a group under composition of these symmetries, $\circ$.

- ~~G_R~~ $\circ$ is used on G_R (see the coming Cayley table). as the composition of two symmetries is a symmetry.

$\circ$ is associative on G_R since composition is always associative.

The <u>identity</u> in $(G_R, \circ)$ is e, the <u>identity function</u> $\mathbb{R}^2 \to \mathbb{R}^2$ leaves all other functions unchanged when composed with e.

$(G_R, \circ)$ contains all the ~~inverses~~ inverses since each symmetry is ~~self~~ self-inverse.

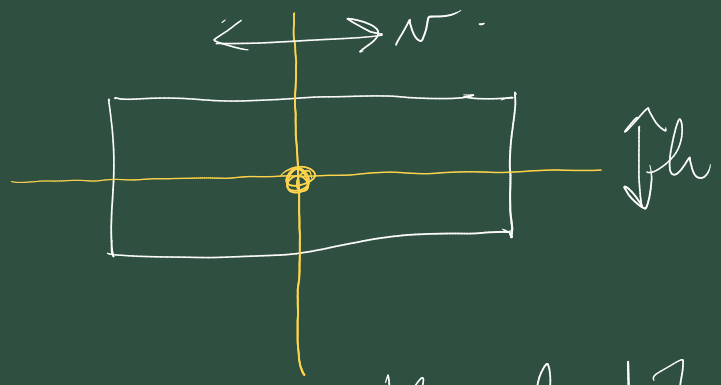
Therefore $(G_R, \circ)$ is a group.

Its Cayley table is

G_R

	e	r	s_1	s_2
e	e	r	s_1	s_2
r	r	e	s_2	s_1
s_1	s_1	s_2	e	r
s_2	s_2	s_1	r	e

Let T be a non-square rectangle



T has four symmetries.

e, r, s_1, s_2 where e is the identity symmetry, r is again a half-rotation and h, v are the reflections in the lines bisecting opposite sides

The Cayley table for $G_T = \{e, r, h, v\}$ is

	e	r	h	v
e	e	r	h	v
r	r	e	v	h
h	h	v	e	r
v	v	h	r	e

G_R and G_T are equivalent as groups as their tables have exactly the same structure under the mapping (isomorphism)

$$e \longmapsto e$$
$$r \longmapsto r$$
$$s_1 \longmapsto h$$
$$s_2 \longmapsto v.$$

Q4(a). Consider

$$20x \equiv 10 \pmod{65}.$$

Let $d = \gcd(20, 65) =$

$$\gcd(2^2 \cdot 5, 5 \cdot 13)$$
$$= 5.$$

Notice $5 \mid 10$, so there are solutions, namely the five different congruence classes modulo 65 represented

by $\boxed{x_i \equiv t + i \cdot 13,}$ $i = 0, \ldots, 4$

where t is the unique solution to the reduced congruence.

$$5x \equiv 2 \pmod{13}.$$

$$\Rightarrow t \equiv 5^{-1} \cdot 2 \pmod{13}.$$

$$\equiv 8 \cdot 2 \pmod{13}$$

since $5^{-1} \equiv 8 \pmod{13}$

as $5 \cdot 8 = 40 = 39 + 1$
$= 3 \cdot 13 + 1$

$$\equiv 16 \pmod{13}$$

$$\equiv 3.$$

So the solutions to the original congruence are

$$x_i = 3 + i \cdot 13, \quad i = 0, \ldots, 4.$$

$$\equiv 3, 16, 29, 42, 55 \pmod{65}$$

(b)

$$x \equiv 5 \pmod 7$$
$$x \equiv 3 \pmod{11}$$
$$x \equiv 11 \pmod{19}.$$

The CRT says this system has unique solution modulo M,

$$M = m_1 m_2 m_3 = 7 \cdot 11 \cdot 19 = 1463$$

(since $7, 11, 19$ are pair-wise coprime).

given by

$$x \equiv \sum_{i=1}^{3} b_i M_i M_i' \pmod{1463}$$

where $b_i = 5, 3, 11$,

$M_1 = 11 \cdot 19 = 209$, $M_2 = 7 \cdot 19 = 133$

$M_3 = 7 \cdot 11 = 77$

$$\begin{array}{r} 29 \\ 7\overline{)209} \end{array} \quad r\ 6.$$

$$M_1' \equiv (M_1)^{-1} \bmod m,$$
$$\equiv (209)^{-1} \pmod 7$$
$$\equiv 6^{-1} \pmod 7. \quad , \ 6 \equiv -1 \pmod 7$$
$$\equiv 6$$

$$M_2' \equiv (M_2)^{-1} \pmod{m_2}$$

$$11\overline{)133}^{\,12 \cdot r1}$$

$$\equiv 133^{-1} \pmod{11}$$

$$\equiv 1^{-1} \pmod{11}$$

$$\equiv 1$$

$$M_3' \equiv (M_3)^{-1} \bmod m_3$$

$$\equiv 77^{-1} \bmod 19$$

$$\equiv 1^{-1} \pmod{19}$$

$$\equiv 1 \pmod{19}.$$

So the required solution is

$$x = b_1 M_1 M_1' + b_2 M_2 M_2' + b_3 M_3 M_3' \pmod{1463}$$

$$\equiv 5 \cdot 209 \cdot 6 + 3 \cdot 133 \cdot 1 + 11 \cdot 77 \cdot 1$$

$$\equiv 7516 \pmod{1463}$$

$$\equiv \underline{\underline{201}} \pmod{1463}.$$

Checking.

$$7 \overline{)209} \quad 28 \text{ r } 5$$

$$11 \overline{)209} \quad 18 \text{ r } 3$$

$$19 \overline{)209} \quad 10 \text{ r } 11$$

$$\begin{array}{r} 569 \\ 373 \\ \hline 196 \end{array}$$

Q (c) Solutions exist $\iff (373|569) = +1$

$$(373|569)$$

$$= (\underline{569}|\underline{373}), \quad 569 \equiv 1 \bmod 4 \text{ by quadratic reciprocity}$$

$$= (196 \mid 373), \text{ by modular reduction.}$$

$$= (2^2 \cdot 7^2 \mid 373)$$

$$= (2|373)^2 \, (7|373)^2.$$

$= 1$, as each Legendre symbol is ± 1 and then squared.

Hence there are solutions to

$$x^2 \equiv 373 \pmod{569}.$$